Calculus of Variations

Prof. Dr. ADEMIR JOAO CAMARGO

1 THE EULER–LAGRANGE EQUATIONS 3
2 FERMAT'S PRINCIPLE AND THE ACTION PRINCIPLE 4
3 MAUPERTUIS AND THE PRINCIPLE OF LEAST ACTION 6
4 EULER AND THE FORMALIZATION OF LEAST ACTION 6
5 LAGRANGE AND VARIATIONAL THEORY 7
6 ALTERNATIVE FORMS OF THE EULER–LAGRANGE EQUATION 12
7 SEVERAL DEPENDENT VARIABLES 13
8 CONSERVATION OF ENERGY 14
9 CONSERVATION OF MOMENTUM 15
10 HOLONOMIC CONSTRAINTS 16

Identities Involving ∇ in $\mathbb{R}^3$

Operation	Definition	Result
Gradient	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$	Vector
Divergence	$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$	Scalar
Curl	$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix}$	Vector
Laplacian	$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$	Scalar

Identity	**Formula**
Divergence of a gradient	$\nabla \cdot (\nabla f) = \nabla^2 f$
Curl of a gradient	$\nabla \times (\nabla f) = \mathbf{0}$
Divergence of a curl	$\nabla \cdot (\nabla \times \mathbf{F}) = 0$
Curl of a curl	$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$
Gradient of a product of scalar fields	$\nabla(fg) = f\nabla g + g\nabla f$
Divergence of a scalar–vector product	$\nabla \cdot (f\mathbf{F}) = f(\nabla \cdot \mathbf{F}) + \nabla f \cdot \mathbf{F}$
Curl of a scalar–vector product	$\nabla \times (f\mathbf{F}) = f(\nabla \times \mathbf{F}) + \nabla f \times \mathbf{F}$

Operation	Definition	Result
Divergence of a cross product		$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G})$
Curl of a cross product		$\nabla \times (\mathbf{F} \times \mathbf{G}) = (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F})$

These identities hold for sufficiently smooth scalar and vector fields in Cartesian coordinates. The vector Laplacian acts componentwise. Equality of mixed partial derivatives underlies the identities involving the curl of a gradient and the divergence of a curl.

The calculus of variations is a fundamental branch of mathematics concerned with extrema: maxima and minima of functionals. Broadly speaking, it seeks the conditions under which a **definite integral** that depends on an unknown function attains a minimum or maximum. The usual strategy is to **express the physical or geometric problem as a functional** and then find a function that makes it stationary. Stationarity is a necessary condition for an interior extremum; additional analysis is generally required to distinguish a minimum, a maximum, and a stationary point that is neither.

To illustrate, consider two distinct points in a plane whose horizontal coordinates are x_i and x_f. Infinitely many curves of the form $y = y(x)$ can connect them, with their endpoint heights fixed. The classical problem is to identify **the curve of shortest length** between these points. The arc-length element ds associated with an infinitesimal displacement along the curve is

$$ds = \sqrt{dx^2 + dy^2}. \qquad 1$$

Thus, the total path length can be written as the functional

$$I[y'(x)] = \int_{x_i}^{x_f} ds = \int_{x_i}^{x_f} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_{x_i}^{x_f} \sqrt{1 + y'^2}\, dx. \qquad 2$$

Finding the shortest curve therefore amounts to minimizing the functional $I[y'(x)]$. Although geometric, this problem has the characteristic structure of problems addressed by the calculus of variations.

Another classic example is the brachistochrone problem, originally posed by Johann Bernoulli in the seventeenth century. Consider a particle, such as a small iron bead sliding without friction, that moves under gravity from a higher initial position at x_i to a lower final position at x_f. Assume uniform gravity, release from rest, and no rolling or dissipative effects. Which curve connecting the two points gives the shortest descent time? The total time is

$$T = \int dt$$

where the infinitesimal time interval is

$$dt = \frac{ds}{\mathrm{v}}$$

The speed can be obtained directly from conservation of mechanical energy:

$$mgh = mgy + \frac{1}{2}m\mathrm{v}^2,$$

which gives

$$\mathrm{v} = \sqrt{2g(h-y)},$$

Here, the initial height is h, and $y = y(x)$ *is the particle's height along the path. Substituting* ds *and the speed into the time integral gives*

$$T = \int_{x_i}^{x_f} \frac{ds}{\mathrm{v}} = \int_{x_i}^{x_f} \frac{\sqrt{1+y'^2}}{\sqrt{2g(h-y)}}dx. \qquad 3$$

The total descent time is therefore expressed by the functional

$$T[y(x), y'(x)] = \int_{x_i}^{x_f} \sqrt{\frac{1+y'^2}{2g(h-y)}}dx \qquad 4$$

Again, the physical problem reduces to minimizing a functional that represents travel time. These examples illustrate questions that the calculus of variations can address: What is the shortest distance between two points in a plane? What is the shortest path between two points on a curved surface, such as a sphere? Which closed plane curve encloses the largest area for a prescribed perimeter? Which shape encloses the largest volume for a prescribed surface area? The stated constraints are essential: without them, the latter two maximization problems have no finite solution.

The power of the calculus of variations lies precisely in this generality: a wide range of physical, geometric, and quantum problems can be formulated as a search for the function that minimizes or maximizes a functional. This perspective will be essential in subsequent chapters, particularly in the variational formulation of quantum mechanics and density functional theory (DFT).

1 The Euler–Lagrange Equations

The modern formulation of the calculus of variations has its roots in an apparently simple question: Which path permits the fastest descent between two points, A and B? This problem, now known as the brachistochrone, immediately reveals the subtlety of finding stationary functionals. At first, intuition suggests that the fastest path should be the shortest: a straight line. A closer physical examination shows why this need not be the best choice. If the path initially curves more steeply downward, the particle gains speed more rapidly under gravity. Although it travels farther, it moves faster over much of the path. The fundamental question, therefore, is which curve best balances the gain in speed against path length.

In the seventeenth century, Galileo believed that the solution was a circular arc. His comparisons of circular and polygonal descent paths did not establish the true path of least time. A definitive answer required a deeper mathematical formulation.

Nearly six decades later, Johann Bernoulli turned the problem into an international challenge. In June 1696, he published an invitation in Acta Eruditorum to Europe's leading mathematicians, asking them to submit solutions within six months. His friend Leibniz later extended the deadline, which helped spread the challenge more widely.

In January 1697, Isaac Newton submitted an anonymous solution for publication in Philosophical Transactions. According to tradition, Bernoulli recognized its author and declared, "I recognize the lion by his claw." Newton's result was elegant and precise, while Johann Bernoulli's own solution was particularly striking: he used a geometric argument that anticipated the variational reasoning underlying the Euler–Lagrange equations.

This episode helped initiate the systematic development of the calculus of variations. The search for the path of least time led to equations that now underlie Lagrangian classical mechanics and much of modern theoretical physics, including density functional theory. The following discussion shows how this problem motivated the variational method and the derivation of the Euler–Lagrange equations, which provide necessary conditions for a functional to be stationary.

2 Fermat's Principle and the Action Principle

To develop his solution to the brachistochrone problem, Johann Bernoulli drew on an old question about the nature of light. As early as the first century CE, thinkers such as Hero of Alexandria asked how light selects its path between two points. In a homogeneous medium such as air, light travels along straight lines. For reflection from a plane surface, the shortest reflected path between fixed points has equal angles of incidence and reflection, with both angles measured from the normal surface.

The situation changes when light crosses the boundary between two media, such as air and water. Its path bends through refraction rather than following the shortest geometric route. The familiar appearance of a pencil "broken" at the water's surface illustrates this behavior. What principle determines the deflection?

During the seventeenth century, experiments established that the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant characteristic of the two media:

$$\frac{\sin \theta_1}{\sin \theta_2} = n.$$

This relation, now known as Snell's law, was well established, but its conceptual justification remained unclear until Fermat developed his principle of least time in the mid-seventeenth century. Fermat, a magistrate by profession and a mathematician by vocation, reconsidered Hero's question: perhaps light minimizes travel time rather than distance. Testing this idea required varying the point at which the path crosses the interface and calculating the travel time for each path. This was a classical variational problem before the calculus of variations had become a formal discipline.

Fermat subsequently showed that Snell's law follows naturally by requiring the total travel time to be stationary; for this simple interface problem, the stationary path is a minimum. The constant n is the ratio of the light speeds in the two media:

$$n = \frac{\mathrm{v}_1}{\mathrm{v}_2}$$

and the law can be rewritten as

$$\frac{\sin\theta_1}{\mathrm{v}_1} = \frac{\sin\theta_2}{\mathrm{v}_2}.$$

Fermat regarded this result as an extraordinary and unexpected discovery. It suggested that an optical law could follow from an optimization principle: the principle of least time in this setting. More generally, Fermat's principle requires stationary optical travel time, which need not always be a minimum.

Familiar with Fermat's work, Bernoulli realized that he could transform the mechanical brachistochrone problem into an optical one. Instead of a particle accelerated by gravity, he considered a light ray passing through layers with a continuously varying refractive index. If each interface obeyed Snell's law, the limit of infinitely thin layers would produce a smooth curve: the desired path.

The crucial question was how the light speed should vary across the layers to reproduce gravitational acceleration. For a particle released from rest, conservation of energy gives

$$\frac{1}{2}m\mathrm{v}^2 = mgy \quad \Rightarrow \quad \mathrm{v} = \sqrt{2gy}$$

Thus, the speed is proportional to $\sqrt{y}$, where y now measures the downward vertical displacement from the initial point, rather than the upward height used in Equations (3) and (4). Substituting this dependence into the generalized form of Snell's law gives

$$\frac{\sin\theta_i}{\sqrt{y_i}} = k.$$

This relation holds throughout the continuously stratified medium. With the angle measured from the vertical, it is equivalent to the differential equation of a cycloid, the curve traced by a point on the edge of a rolling wheel. Thus, the brachistochrone is an arc of an inverted cycloid, rather than a straight line or a circular arc. The constant absorbs the factor associated with gravitational acceleration.

An inverted cycloid also has the remarkable property that, under the same ideal assumptions, a particle released from rest reaches its lowest point at the same time, regardless of its starting point on the same arch. It is therefore also called a tautochrone, or “equal-time” curve. This property concerns descent to the lowest point, not travel to an arbitrary endpoint.

3 Maupertuis and the Principle of Least Action

Dissatisfied with the predominance of Cartesian thought in France and interested in developments in Leibnizian calculus, Maupertuis sought Johann Bernoulli's guidance in Switzerland. Their relationship developed into a lasting friendship, and Maupertuis later died at the home of Johann's son Daniel Bernoulli. Building on this theoretical background, he sought a unifying principle for light and matter. Observing analogous behavior in nature, he proposed that a fundamental quantity, *action*, is minimized:

$$S = mvs,$$

the product of mass, speed, and distance traveled. For motion composed of several segments, the action would be

$$S = \sum_i m\, v_i s_i.$$

In 1744, Maupertuis famously described action as *Nature's expenditure*, which Nature strives to make as small as possible.

Despite its visionary character, the proposal attracted substantial criticism. Samuel König accused Maupertuis of conceptual error and of plagiarizing Leibniz, and Voltaire, once his friend, sharply attacked his work. Part of the difficulty was the lack of a rigorous mathematical foundation: why should nature minimize mass times speed times distance, and under what conditions should the principle hold?

4 Euler and the Formalization of Least Action

Leonhard Euler, also a student of Johann Bernoulli, became a major defender of Maupertuis. Euler replaced the discrete expression with a continuous integral:

$$\sum m\, vs \to \int m\, v ds.$$

This made it possible to apply the principle to systems in which speed and direction vary continuously, such as planetary orbits. Mathematical tools were still needed to handle infinitesimal variations of entire paths. Euler developed a new method that directly anticipated the modern calculus of variations.

This formulation identifies two essential requirements for Maupertuis' principle in its usual conservative form: total energy is conserved along each admissible path, and all paths being compared have the same total energy. More precisely, the reduced action is stationary; it need not be a minimum.

5 Lagrange and Variational Theory

Joseph-Louis Lagrange made a decisive contribution while still a young mathematician. In his correspondence with Euler in the 1750s, he introduced an analytic variational method that helped establish the Lagrangian formulation of classical mechanics. Starting from Euler's form of Maupertuis' principle for a particle of constant mass, we may write

$$\delta \int m\mathrm{v}\, ds \; = 0 \Rightarrow \delta \int m\mathrm{v}^2 dt \; = 0 \Rightarrow \delta \int \left(\frac{m\mathrm{v}^2}{2} + \frac{m\mathrm{v}^2}{2} \right) dt = 0.$$

For a conservative system, total energy is the sum of kinetic and potential energy:

$$E = \frac{m\mathrm{v}^2}{2} + U \Longrightarrow \frac{m\mathrm{v}^2}{2} = E - U$$

Substituting this relation into the preceding expression gives

$$\delta \int \left(\frac{m\mathrm{v}^2}{2} + E - U \right) dt = 0.$$

Separate the integral into two terms:

$$\delta \int \left(\frac{m\mathrm{v}^2}{2} - U \right) dt + \delta \int E\, dt = 0.$$

Because energy is constant along each path, its integral is the energy multiplied by the elapsed time:

$$\delta \int_{t_i}^{t_f} \left(\frac{m\mathrm{v}^2}{2} - U \right) \, dt + \delta[E(t_f - t_i)] = 0.$$

The product rule for variations gives two contributions. For the fixed-energy family of paths considered by Maupertuis, the energy variation vanishes:

$$\delta[E(t_f - t_i)] = (t_f - t_i)\delta\, E + E\delta(t_f - t_i) = E\delta(t_f - t_i).$$

Rearranging gives the relation between the variation of the time action and the variation of the elapsed time:

$$\delta \int_{t_i}^{t_f} \left(\frac{m\mathrm{v}^2}{2} - U \right) \, dt = -E\delta(t_f - t_i).$$

If the initial and final times are also fixed, the elapsed-time variation vanishes, leaving

$$\delta \int_{t_i}^{t_f} \left(\frac{m\mathrm{v}^2}{2} - U \right) \, dt = 0.$$

This calculation connects the two action expressions under the stated restrictions. It does not, by itself, justify extending a fixed-energy variational principle to all paths with fixed endpoint times. Hamilton's principle makes that broader statement independently: *the time action is stationary for arbitrary sufficiently smooth variations that vanish at fixed spacetime endpoints, without requiring the trial paths to have equal energies.*

In the modern formulation associated with William Rowan Hamilton's work of 1834, the integrand is the Lagrangian:

$$\delta \int_{t_i}^{t_f} L(q, \dot{q}, t)\, dt = 0,$$

where q is a generalized coordinate and L is the Lagrangian, given here for a single Cartesian degree of freedom with constant mass by

$$L = \frac{m\dot{q}^2}{2} - U(q).$$

Hamilton's principle determines the system's time evolution, whereas Maupertuis' principle determines the geometric path at fixed energy. In Hamilton's principle, the initial and final positions and times are fixed, while trial paths may have different energies and need not conserve energy individually. In Maupertuis' principle, the trial paths have a common conserved energy, while the travel time may vary. The term "*principle of least action*" is conventional: the general requirement is stationarity.

Thus, the physical trajectory satisfies the stationary-action condition

$$\delta\, S = \delta \int_{t_i}^{t_f} (T - U) dt = 0.$$

Our next task is to obtain the equations that make a functional stationary. Given two fixed endpoints in a plane, with horizontal coordinates x_i and x_f, infinitely many curves $y = y(x)$ pass through them. We seek the curve $y = y(x)$ that makes the following integral stationary:

$$I = \int_{x_i}^{x_f} F(x, y, y')\, dx.$$

To use the techniques of ordinary calculus, introduce a family of functions $Y(x, \epsilon)$, defined by

$$Y(x, \epsilon) = y(x) + \epsilon\eta(x), \tag{5}$$

Here, ϵ is a small real parameter and $\eta(x)$ is an arbitrary sufficiently smooth function that vanishes at the endpoints: $\eta(x_i) = \eta(x_f) = 0$. Consequently, $Y(x_i, \epsilon) = y(x_i)$ and $Y(x_f, \epsilon) = y(x_f)$. Holding $\eta(x)$ fixed and varying ϵ generates a family of curves $Y(x, \epsilon)$. Thus, $Y(x, \epsilon)$ also depends on ϵ, and the functional evaluated on each curve depends on ϵ:

$$I(\epsilon) = \int_{x_i}^{x_f} F(x, Y, Y')dx. \tag{6}$$

Y' is given by

$$Y'(x, \epsilon) = y'(x) + \epsilon\eta'(x).$$

We write I as a function of ϵ alone because the integration over x has already been performed in I; the dependence on the chosen variation function is understood. A necessary condition for stationarity of Equation (6) is

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = 0 \tag{7}$$

More explicitly,

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \left[\frac{d}{d\epsilon}\int_{x_i}^{x_f} F(x,Y,Y')dx\right]_{\epsilon=0} = \int_{x_i}^{x_f}\left[\frac{d}{d\epsilon}F(x,Y,Y')\right]_{\epsilon=0} dx = 0 \tag{8}$$

Assume the integrand and variation are sufficiently smooth on the finite integration interval so that differentiation under the integral sign is valid. The endpoints do not depend on ϵ , and integration is over x. Applying the chain rule to $F(x,Y,Y')$ with respect to ϵ gives

$$\frac{dF(x,Y,Y')}{d\epsilon} = \frac{\partial F}{\partial x}\underbrace{\frac{\partial x}{\partial \epsilon}}_{=0} + \frac{\partial F}{\partial Y}\frac{\partial Y}{\partial \epsilon} + \frac{\partial F}{\partial Y'}\frac{\partial Y'}{\partial \epsilon} = \frac{\partial F}{\partial Y}\eta(x) + \frac{\partial F}{\partial Y'}\eta'^{(x)} \tag{9}$$

The derivative $\partial x/\partial\epsilon = 0$ because x is independent of ϵ. Substituting Equation (9) into Equation (8) gives

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \left[\int_{x_i}^{x_f}\left(\frac{\partial F}{\partial Y}\eta(x) + \frac{\partial F}{\partial Y'}\eta'(x)\right)dx\right]_{\epsilon=0} = 0 \tag{10}$$

Evaluating Equation (10) at $\epsilon = 0$ gives

$$\int_{x_i}^{x_f}\left(\frac{\partial F}{\partial y}\eta(x) + \frac{\partial F}{\partial y'}\eta'^{(x)}\right)dx = 0. \tag{11}$$

This is the weak variational condition: it contains both the test function $\eta(x)$ and its derivative $\eta'(x)$. To remove the derivative of the test function, integrate by parts:

$$\int_{x_i}^{x_f}\frac{\partial F}{\partial y'}\eta'(x)\,dx = \underbrace{\left[\frac{\partial F}{\partial y'}\eta(x)\right]_{x_i}^{x_f}}_{=0} - \int_{x_i}^{x_f}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta(x)\,dx. \tag{12}$$

The boundary term in Equation (12) vanishes because $\eta(x_i) = \eta(x_f) = 0$. Substituting Equation (12) into Equation (11) gives

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \int_{x_i}^{x_f}\left(\frac{\partial F}{\partial y}\eta(x) - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta(x)\right)dx = 0. \tag{13}$$

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \int_{x_i}^{x_f}\left(\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right)\eta(x)dx = 0. \tag{14}$$

Since $\eta(x)$ is arbitrary in the interior, the fundamental lemma of the calculus of variations implies

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0. \tag{15}$$

Equation (15) is the Euler–Lagrange equation, a central result of analytical mechanics. We now relate the first variation δI to the derivative

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} \tag{16}$$

Expand the integrand $F[x,y,y']$ in the Taylor series with respect to its second and third arguments:

$$F[x, y+\delta y, y'+\delta y'] = F[x, y, y'] + \frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y' + O(\delta^2)$$

$$\underbrace{F[x, y+\delta y, y'+\delta y'] - F[x, y, y']}_{\Delta F} = \frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y' + O(\delta^2) \tag{17}$$

The first variation of the integrand F is its linear part:

$$\boxed{\delta F = \frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'} \tag{18}$$

Integrating the expansion in Equation (17) gives

$$\begin{aligned} &\underbrace{\int_{x_i}^{x_f} F[x, y+\delta y, y'+\delta y']dx - \int_{x_i}^{x_f} F[x, y, y']dx}_{\delta^{(T)}I} \\ &= \underbrace{\int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'\right) dx}_{\delta^{(1)}I} + O(\delta^2), \end{aligned} \tag{19}$$

Here, $\delta^{(T)}I$ and $\delta^{(1)}I$ denote the total change and first variation of the functional whose integrand is $F[x, y, y']$, respectively. Neglecting terms of order $O(\delta^2)$ retains only the part of Equation (19) that is linear in the variations:

$$\delta^{(1)}I = \int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'\right) dx, \tag{20}$$

The notation $\delta^{(1)}I$ emphasizes that only the first variation is retained. We shall write this simply as

$$\delta I = \int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'\right) dx \tag{21}$$

This is the weak form, involving both δy and $\delta y'$. Integrate the second term of Equation (21) by parts. The boundary condition $\delta y(x_i) = \delta y(x_f) = 0$ removes the boundary term, since the endpoint values of $y(x)$ at x_i and x_f are fixed:

$$\int_{x_i}^{x_f} \frac{\partial F}{\partial y'}\delta y' dx = \underbrace{\frac{\partial F}{\partial y'}\delta y\Big|_{x_i}^{x_f}}_{=0} - \int_{x_i}^{x_f} \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\delta y dx = -\int_{x_i}^{x_f} \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\delta y dx$$

Substituting this result into Equation (21) gives

$$\delta I = \int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y}\delta y - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\delta y\right) dx = \int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right)\delta y dx \tag{22}$$

Requiring the first variation to vanish for every admissible δy, the fundamental lemma yields the necessary stationarity condition for the functional with integrand $F[x, y(x), y'(x)]$:

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0, \tag{23}$$

This is again the Euler–Lagrange equation. Lagrange also obtained the equations of motion through d'Alembert's principle; the variational derivation above follows Euler's approach. Equation (5) gives

$$Y(x,\epsilon) = y(x) + \epsilon\eta(x) \Rightarrow \underbrace{Y(x,\epsilon) - y(x)}_{\delta y} = \epsilon\eta(x) \Rightarrow \boxed{\delta y = \epsilon\eta(x)}$$

Substituting this result into Equation (22) gives

$$\delta I = \int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) \right) \epsilon\eta(x)dx \tag{24}$$

Equations (24) and (14) have the same integral, with an additional factor ϵ in Equation (24). Hence,

$$\delta I = \left[\int_{x_i}^{x_f} \left(\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) \right) \eta(x)dx \right] \epsilon = \left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} \epsilon \tag{1.25}$$

Equation (1.25) shows that $y(x)$ is a stationary function for the functional with integrand $F[x, y, y']$ when $\delta I = 0$ for every admissible variation, or equivalently when

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = 0.$$

The variation operator δ commutes with the derivative operator d/dx for variations at a fixed independent variable. Differentiating Equation (5) gives

$$Y'(x,\epsilon) = y'(x) + \epsilon\eta'(x) \Rightarrow \underbrace{Y'(x,\epsilon) - y'(x)}_{\delta y'} = \epsilon\eta'(x)$$

$$\boxed{\delta y' = Y'(x,\epsilon) - y'(x) = \epsilon\eta'(x)} \tag{26}$$

$$\delta y' = \delta\left(\frac{dy}{dx}\right) = Y'(x,\epsilon) - y'(x) = \frac{dY}{dx} - \frac{dy}{dx} = \frac{d}{dx}\underbrace{(Y - y)}_{\delta y} = \frac{d}{dx}(\delta y) \tag{27}$$

Therefore,

$$\delta\left(\frac{dy}{dx}\right) = \frac{d}{dx}(\delta\, y).$$

For fixed integration limits and sufficiently smooth variations, δ also commutes with integration:

$$\delta \int y(x)dx = \int \delta y(x)dx. \tag{1.28}$$

Exercises

Find the equation of the curve that minimizes the distance between two points in a plane.

Solution

$$I = \int_{x_i}^{x_f} \sqrt{1 + y'^2}\, dx.$$

In this case,

$$F[x, y(x), y'(\mathrm{x})] = \sqrt{1 + y'^2}.$$

Applying the Euler–Lagrange equation gives

$$\frac{\partial F}{\partial y} = 0, \qquad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2}}, \qquad \frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2}}\right) = 0.$$

The map from the slope to the displayed ratio is strictly increasing, so a constant ratio implies a constant slope. The endpoint values determine the slope and intercept. The straight segment is globally shortest by the triangle inequality.

$$\frac{y'}{\sqrt{1 + (y')^2}} = k \Rightarrow\ y' = c \Rightarrow\ y(x) = cx + b.$$

6 Alternative Forms of the Euler–Lagrange Equation

Two useful first integrals follow from the Euler–Lagrange equation

$$\frac{\partial F}{\partial \mathrm{y}} - \frac{d}{dx}\left(\frac{\partial F}{\partial \mathrm{y}'}\right) = 0. \tag{29}$$

They apply when the integrand has no explicit dependence on x or no explicit dependence on y. These conditions concern the integrand, rather than the appearance of the resulting Euler–Lagrange equation. If the integrand does not depend explicitly on y, then

$$\frac{d}{dx}\left(\frac{\partial F}{\partial \mathrm{y}'}\right) = 0.$$

and therefore

$$\frac{\partial F}{\partial \mathrm{y}'} = constant. \tag{30}$$

If F does not depend explicitly on x, we obtain the *Beltrami identity*. Multiply the Euler–Lagrange equation by y':

$$y'\frac{\partial F}{\partial y} - y'\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0. \tag{31}$$

Taking the total derivative of $F\big(x, y(x), y'(x)\big)$ with respect to x gives

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} + \frac{\partial F}{\partial y'}\frac{dy'}{dx} = \frac{\partial F}{\partial x} + y'\frac{\partial F}{\partial y} + y''\frac{\partial F}{\partial y'}. \tag{32}$$

Solve Equation (32) for its second term on the right and substitute into Equation (31). The last two terms are combined by the product rule, with a minus sign applying to both:

$$\frac{dF}{dx} - \frac{\partial F}{\partial x} - y'' \frac{\partial F}{\partial y'} - y' \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0.$$
$$\frac{dF}{dx} - [y'' \frac{\partial\, F}{\partial\, y'} + y' \frac{d}{dx}(\frac{\partial\, F}{\partial\, y'})] = \frac{\partial\, F}{\partial\, x}.$$
$$\frac{dF}{dx} - \frac{d}{dx}\left(y' \frac{\partial F}{\partial y'}\right) = \frac{\partial F}{\partial x}$$
$$\frac{d}{dx}\left(F - y' \frac{\partial F}{\partial y'}\right) = \frac{\partial F}{\partial x} \tag{33}$$

If F has no explicit dependence on x, the right-hand side of Equation (33) vanishes:

$$\frac{d}{dx}\left(F - y' \frac{\partial F}{\partial y'}\right) = 0$$
$$F - y' \frac{\partial F}{\partial y'} = constant. \tag{34}$$

Exercise. Use the Beltrami identity to find stationary curves for the functionals in Equations (2) and (4), and examine which solutions minimize them under the stated endpoint conditions.

7 Several Dependent Variables

The Euler–Lagrange equation derived above involves one dependent variable: $F = F(x, y, y')$. Now consider an integrand with several dependent variables: $F = F(x, y_1, y_2, \cdots, y_n, y_1', y_2', \cdots, y_n')$. The Euler–Lagrange equations are

$$\frac{\partial F}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_i'}\right) = 0, \quad i = 1, \cdots, n. \tag{35}$$

To derive Equation (35), seek functions $y_1(x), y_2(x), \cdots, y_n(x)$ that make the following integral stationary:

$$I = \int_{x_i}^{x_f} F(x, y_1, \ldots, y_n, y_1', \ldots, y_n')\, dx. \tag{36}$$

The procedure parallels the derivation for one dependent variable. Infinitely many paths connect fixed endpoint values at x_i and x_f. Neighboring paths are described by

$$Y_i(x, \varepsilon) = y_i(x) + \varepsilon \eta_i(x), \qquad \eta_i(x_i) = \eta_i(x_f) = 0. \tag{37}$$

Each dependent variable has its own independent variation function, vanishing at both endpoints. Expanding the functional in Equation (36) in a Maclaurin series in the variation parameter gives

$$I(\epsilon) = I(0) + \left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} \epsilon + O(\epsilon^2).$$

Neglecting terms of order $O(\epsilon^2)$, the first variation is

$$\delta I = I(\epsilon) - I(0) = \left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} \epsilon.$$

Requiring $\delta I = 0$ for every admissible direction is equivalent to

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = 0,$$

because the coefficient of the arbitrary first-order parameter ϵ must vanish. Substituting the integral expression for I and differentiating under the integral sign gives

$$\frac{dI}{d\varepsilon}|_{\varepsilon=0} = \int_{x_i}^{x_f} \frac{d}{d\varepsilon} F(x, Y_1, \dots, Y_n, Y_1', \dots, Y_n')|_{\varepsilon=0}\, dx = 0.$$

$$\int_{x_i}^{x_f} \sum_{i=1}^{n} \left(\frac{\partial F}{\partial Y_i}\frac{\partial Y_i}{\partial \varepsilon} + \frac{\partial F}{\partial Y_i'}\frac{\partial Y_i'}{\partial \varepsilon}\right)|_{\varepsilon=0}\, dx = 0.$$

Integrate the derivative term in the sum by parts and use Equation (37), with each variation function vanishing at the endpoints:

$$\left[\int_{x_i}^{x_f} \sum_i \left[\frac{\partial F}{\partial Y_i}\eta_i(x) - \frac{d}{dx}\left(\frac{\partial F}{\partial Y_i'}\right)\eta_i(x)\right] dx\right]_{\epsilon=0} = 0$$

$$\left[\int_{x_i}^{x_f} \sum_i \left[\frac{\partial F}{\partial Y_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial Y_i'}\right)\right]\eta_i(x)\, dx\right]_{\epsilon=0} = 0.$$

Since the variation functions are independent, the fundamental lemma gives Equation (35):

$$\frac{\partial F}{\partial \mathrm{y}_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial \mathrm{y}_i'}\right) = 0, \quad i = 1, \cdots, n.$$

Thus, when a functional depends on several functions, there is one Euler–Lagrange equation for each dependent function. Together these equations provide necessary conditions for stationarity; further analysis determines whether an extremum is obtained.

8 Conservation of Energy

Energy conservation follows from time-translation symmetry when the Lagrangian has no explicit time dependence. This means the laws governing the system do not depend on the choice of time origin; it does not mean every repetition of an experiment yields the same numerical energy regardless of initial conditions. Consider an infinitesimal time translation by ε:

$$t \to t + \varepsilon$$

Along a trajectory, the value of the Lagrangian at the shifted time has the first-order expansion

$$L(t + \varepsilon) = L(t) + \varepsilon \frac{dL}{dt} + O(\varepsilon^2).$$

This expansion describes the total change along the trajectory, not explicit time dependence alone.

For an autonomous Lagrangian, the explicit partial derivative with respect to time vanishes. Taking the total derivative of the Lagrangian then gives

$$\frac{dL}{dt} = \frac{\partial L}{\partial t} + \frac{\partial L}{\partial x}\dot{x} + \frac{\partial L}{\partial \dot{x}}\ddot{x}.$$

$$\frac{\partial L}{\partial t} = 0 \Rightarrow \frac{dL}{dt} = \frac{\partial L}{\partial x}\dot{x} + \frac{\partial L}{\partial \dot{x}}\ddot{x}.$$

Using the Euler–Lagrange equation gives

$$\frac{dL}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\dot{x} + \frac{\partial L}{\partial \dot{x}}\ddot{x}.$$

The product rule combines the two terms on the right into a total derivative. Thus,

$$\frac{d}{dt}\left(\dot{x}\frac{\partial L}{\partial \dot{x}} - L\right) = 0.$$

For a constant-mass particle with a time-independent potential and the usual kinetic-minus-potential Lagrangian,

$$\frac{d}{dt}\left(m\dot{x}^2 - \frac{m\dot{x}^2}{2} + U\right) = \frac{d}{dt}\left(\frac{m\dot{x}^2}{2} + U\right) = \frac{dE}{dt} = 0.$$

Therefore, $E = constant$. More generally, time-translation invariance conserves the corresponding energy function; its identification with kinetic plus potential energy uses the particular Lagrangian assumed here.

9 Conservation of Momentum

Conservation of linear momentum follows from spatial-translation invariance. Consider a constant infinitesimal translation of the position x:

$x \rightarrow x + \varepsilon$

$\dot{x} \rightarrow \dot{x}$

The velocity is unchanged because the translation is constant. Translation invariance is a property of the Lagrangian to be assumed or checked for the system; it does not follow from the Euler–Lagrange equation alone.

$$\frac{d}{dt}(x + \varepsilon) = \dot{x}, \qquad \frac{d\varepsilon}{dt} = 0.$$

$\therefore \quad L(t, x + \varepsilon, \dot{x}) = L(t, x, \dot{x})$

The change in the Lagrangian under this translation is

$$\delta L = L(t, x + \varepsilon, \dot{x}) - L(t, x, \dot{x})$$

Expanding the first term on the right in a Taylor series gives

$$\delta L = L(t, x, \dot{x}) + \frac{\partial L}{\partial x}\varepsilon + O(\varepsilon^2) - L(t, x, \dot{x})$$

$$\delta L = \frac{\partial L}{\partial x}\varepsilon$$

Neglect terms of order $O(\varepsilon^2)$. If L is invariant under arbitrary infinitesimal translations in this direction, then

$$\frac{\partial L}{\partial x} = 0.$$

The Euler–Lagrange equation consequently gives conservation of the conjugate momentum:

$$\frac{dp}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x} = 0 \Rightarrow p = \text{constant}.$$

10 Holonomic Constraints

A holonomic constraint is a relation among the coordinates that can be expressed as an equation of the form

$$f(q_1, q_2, \cdots, q_n, t) = 0,$$

where q_i are generalized coordinates. A constraint with no explicit dependence on time t is called scleronomic. Constraints make the coordinate variations dependent. The same mathematics applies to dependent variables in a variational problem. Consider $F(x, y_1, \ldots, y_n, y_1', \ldots, y_n')$, with n dependent variables $y_i(x)$ related by m constraints:

$$f_1(x, y_1, y_2, \cdots, y_n) = 0$$
$$\vdots$$
$$f_m(x, y_1, y_2, \cdots, y_n) = 0.$$

Assume the constraints are smooth and independent, meaning that their Jacobian with respect to the dependent variables has full row rank locally. Then n variables subject to m constraints have $n - m$ local degrees of freedom. One may eliminate constrained variables and work with $n - m$ Euler–Lagrange equations, but this can be cumbersome. An alternative is to retain all n variables and introduce Lagrange multipliers. At fixed independent variable, the linearized constraints are

$$\delta f_1 = \frac{\partial f_1}{\partial y_1}\delta y_1 + \frac{\partial f_1}{\partial y_2}\delta y_2 + \cdots + \frac{\partial f_1}{\partial y_n}\delta y_n = 0$$
$$\vdots$$
$$\vdots$$
$$\delta f_m = \frac{\partial f_m}{\partial y_1}\delta y_1 + \frac{\partial f_m}{\partial y_2}\delta y_2 + \cdots + \frac{\partial f_m}{\partial y_n}\delta y_n = 0.$$

Multiply each constraint variation by an undetermined Lagrange multiplier λ and sum:

$$\lambda_1 \delta f_1 + \lambda_2 \delta f_2 + \cdots + \lambda_m \delta f_m = 0. \tag{38}$$

Since $f_1(x, y_1, y_2, \cdots, y_n) = 0$ holds along every admissible path, $\delta f_i = 0$ for $i = 1 \cdots m$, which explains Equation (38). For a functional, stationarity means that its first variation vanishes for all variations compatible with the constraints:

$$\delta I = \int_{x_i}^{x_f} \sum_{i=1}^{n} \left(\frac{\partial F}{\partial y_i}\delta y_i + \frac{\partial F}{\partial y_i'}\delta y_i'\right) dx = 0.$$

A vanishing integral does not imply that its integrand vanishes pointwise. First integrate the terms containing derivatives of the variations by parts, using fixed endpoint values. Define the Euler–Lagrange residuals by

$$A_i = \frac{\partial F}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_i'}\right), \qquad \delta I = \int_{x_i}^{x_f} \sum_{i=1}^{n} A_i \delta y_i \, dx = 0. \tag{39}$$

The stationarity condition is then a weighted integral of these residuals. Because the admissible variations are constrained, their coefficients cannot yet be set separately to zero. Add the identically zero constraint contribution in Equation (38), integrated over the same interval:

$$\int_{x_i}^{x_f} \sum_{i=1}^{n} \left(A_i + \sum_{a=1}^{m} \lambda_a(x) \frac{\partial f_a}{\partial y_i} \right) \delta y_i \, dx = 0. \tag{40}$$

For regular constraints, the residual vector must lie in the span of the constraint gradients. Thus, there exist multiplier functions that give the augmented stationarity condition

$$A_i + \sum_{a=1}^{m} \lambda_a(x) \frac{\partial f_a}{\partial y_i} = 0, \quad i = 1, \ldots, n.$$

Equivalently, in terms of the original integrand,

$$\frac{\partial F}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_i'}\right) + \sum_{a=1}^{m} \lambda_a(x) \frac{\partial f_a}{\partial y_i} = 0, \quad f_a(x, y) = 0. \tag{41}$$

Here the multipliers are functions of the independent variable because the holonomic constraints hold pointwise along the path. The equations follow from stationarity of the augmented functional with respect to unrestricted variations of the dependent functions and the multipliers. For finite-dimensional optimization, there is no integral and no derivative dependence, so these conditions reduce to

$$\frac{\partial F}{\partial y_i} + \sum_{a=1}^{m} \lambda_a \frac{\partial f_a}{\partial y_i} = 0, \quad i = 1, \ldots, n. \tag{42}$$

In the finite-dimensional case, Equation (42) together with the constraints can be expressed in differential form as

$$\delta F + \lambda_1 \delta f_1 + \lambda_2 \delta f_2 + \cdots + \lambda_m \delta f_m = 0, \tag{43}$$

or, equivalently, taking the variation with respect to the dependent variables while holding the multipliers fixed,

$$\delta(F + \lambda_1 f_1 + \lambda_2 f_2 + \cdots + \lambda_m f_m) = 0, \tag{44}$$

The multipliers are determined together with the stationary coordinates by Equation (42) and the constraint equations. The augmented expression allows unrestricted coordinate variations; the constraints are still imposed by varying with respect to the multipliers. For the general variational problem, use the augmented integrand defined below:

$$\tilde{F} = F + \sum_{a=1}^{m} \lambda_a(x) f_a(x, y).$$

When there is only one constraint, the finite-dimensional condition in Equation (44) becomes

$$\delta(F + \lambda f) = 0. \tag{45}$$

This produces one stationarity equation for each dependent variable:

$$\frac{\partial F}{\partial y_i} + \lambda \frac{\partial f}{\partial y_i} = 0, \quad i = 1, \cdots, n.$$

The constraint supplies one additional equation. Thus, there are as many equations as unknown coordinates and multipliers, but this count does not guarantee existence or uniqueness. Several stationary points may occur, and they must be classified separately. In the variational case, the unknowns are functions and the equations are differential equations supplemented by constraints and boundary conditions.

Example 1. Consider the height function $\Theta(x, y) = x + y$ above the xy plane. Find its maximum and minimum on the circle defined by $(x - 2)^2 + (y - 2)^2 = 1$. Geometrically, this means finding the highest and lowest points of the circle's vertical projection onto the plane that represents the height function.

Solution

$$\Theta(x, y) = x + y$$
$$f(x, y) = (x - 2)^2 + (y - 2)^2 - 1 = 0$$

The augmented stationarity condition

$$\delta(\Theta + \lambda f) = 0,$$

gives the following finite-dimensional equations:

$$\frac{\partial}{\partial x}(\Theta + \lambda f) = 0 \Longrightarrow 1 + 2\lambda(x - 2) = 0$$
$$\frac{\partial}{\partial y}(\Theta + \lambda f) = 0 \Longrightarrow 1 + 2\lambda(y - 2) = 0.$$

The constraint $f = 0$ supplies a third equation:

$$(x - 2)^2 + (y - 2)^2 - 1 = 0 \tag{46}$$

Solving the first two equations for λ and equating the results gives $x = y$. Substituting y for x in Equation (46) gives

$$x = 2 \pm \frac{\sqrt{2}}{2}.$$

Thus, the stationary points of $\Theta(x, y) = x + y$ subject to $(x - 2)^2 + (y - 2)^2 = 1$ are $\left(2 + \frac{\sqrt{2}}{2}, 2 + \frac{\sqrt{2}}{2}\right)$ and $\left(2 - \frac{\sqrt{2}}{2}, 2 - \frac{\sqrt{2}}{2}\right)$. The point with the plus signs is the global maximum, and the point with the minus signs is the global minimum, because the constraint circle is compact and these are its only stationary points.

Example 2. Find the maximum value of $f(x, y) = x^2 y$ subject to x and y lying on the circle $x^2 + y^2 = 3$.